

XINGCHEN SHA

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EDUCATION

Ph.D. in Computer Science Northwestern University, Evanston, United States	Graduating May 2030 (expected)
B.A. in Computer Science Columbia University, New York, United States	Graduating May 2025 GPA: 4.06
B.S. in Computer Science City University of Hong Kong, Kowloon Tong, Hong Kong	Graduating May 2025 GPA: 4.01

EXPERIENCE

Columbia University: Research Assistant • Collaborator in the NSF CSIRO-funded project Fair Sequential Collective Decision Making • Researched automatic algorithms design with LLMs • Researched group-fairness in multi-period facility location problems • Advisor: Haris Aziz, Hau Chan, Toby Walsh, Lirong Xia	October 2023 – Present
City University of Hong Kong: Research Assistant • Researched facility location games with capacity constraints • Researched multi-stage facility location games in dynamic settings • Researched truthful mechanism design with optional preferences • Advisor: Hau Chan, Vincent Chau, Ken Fong, Minming Li, Pinyan Lu, Jialin Zhang	June 2022 – April 2023

ACTIVITY

Columbia University: Presenter & Teaching Assistant Decentralized Finance and Game Theory Seminar	September 2024 – Present
City University of Hong Kong: Spotlight Presenter STEM Carnival cum Student Project Exhibition 2023	June 2023
City University of Hong Kong: Organizer Hong Kong Secondary School Coding Challenge 2023 (HKSC2023)	May 2023 – September 2023
City University of Hong Kong: Presenter CityU Theory Seminar	September 2022 – September 2023
City University of Hong Kong: Organizer Hong Kong Secondary School Coding Challenge 2022 (HKSC2022)	May 2022 – September 2022

SERVICES

City University of Hong Kong: College of Engineering Student Chapter Panel	November 2022 - Present
City University of Hong Kong: Freshman Mentor	September 2022 - December 2022

SELECTED HONORS & AWARDS

GS Honor Society	February 2025
Upsilon Pi Epsilon	September 2024
Dean's List at Columbia University	September 2023 – January 2025
VTech Group of Companies Scholarship	June 2023
Dean's List at City University of Hong Kong	September 2020 – May 2023
The College of Engineering Dean's Scholarship	April 2023
HKSAR Government Scholarship	February 2023

UNDER REVIEW (* ALPHABETICAL ORDER)

Large Language Models for Indivisible Participatory Budgeting

Nguyen Thach, **Xingchen Sha**, Hau Chan

Group Fairness in Multi-period Mobile Facility Location Problems *

Haris Aziz, Hau Chan, **Xingchen Sha**, Toby Walsh, Lirong Xia

PUBLICATIONS (* ALPHABETICAL ORDER)

Mechanism Design for Facility Location Problems with Capacity Constraints in Bounded Location Space

Xingchen Sha, Hau Chan, Vincent Chau, Ken Fong, Minming Li, Wai Lun Lo, ECAI 2025

Strategyproof Mechanism for Two Heterogeneous Facilities with Constant Approximation Ratio *

Minming Li, Pinyan Lu, **Xingchen Sha**, Yuhao Yao, Jialin Zhang, Games and Economic Behavior (#1 specialty journal in game theory)

Facility Location Games with Optional Preferences: A Revisit

Xingchen Sha, Shuyu Bao, Hau Chan, Vincent Chau, Ken Fong, Minming Li, AAAI 2025

Randomized Strategyproof Mechanisms for Multi-stage Facility Location Problem with Capacity Constraints

Ken Fong, **Xingchen Sha**, Hau Chan, Vincent Chau, Wai Lun Lo, IJTCS-FAW 2024

Group Fairness in Multi-period Mobile Facility Location Problems *

Haris Aziz, Hau Chan, **Xingchen Sha**, Toby Walsh, Lirong Xia, AAMAS 2025 (Extended Abstract), ADT 2024 (Abstract)

SKILLS

Programming Languages:

C/C++, Python, Java, SQL, Bash, HTML

Professional Applications:

Latex, Matlab, PyTorch, TensorFlow, Git